COPY-PASTE TO MITIGATE LARGE LANGUAGE MODEL HALLUCINATIONS

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ABSTRACT

Retrieval-Augmented Generation (RAG) is a widely adopted strategy for grounding large language model (LLM) outputs in external evidence, yet hallucinations persist because models often favor parametric memory over supplied context. We identify a fundamental cause: LLMs inherently lean toward abstractive paraphrasing, distancing generated text from the source material. Motivated by an empirical finding on RAGTruth that copying degree is inversely correlated with context-unfaithful hallucinations, we propose **Copy-Paste**—a generation paradigm that treats the direct reuse of contextual fragments as the primary vehicle for faithfulness and attribution. Building on this, we introduce **CopyPasteLLM**, a two-stage pipeline. Stage 1 (Copy-Paste-Prompting) generates high-copying responses via CP-Order, CP-Link, and CP-Refine strategies. Stage 2 builds DPO preference pairs ranked by Elo LLM-as-Judge and trains CopyPasteLLM with only 365 seed samples. CopyPasteLLM achieves up to 92.8% on FaithEval (+12.6 pp over the best baseline) using $\approx 1/50$th the training data, and generalizes strongly to the unseen ConFiQA benchmark (83.6% QA).

1 INTRODUCTION

Large language models (LLMs) have achieved remarkable performance across a wide range of tasks, but their tendency to hallucinate—generating plausible-sounding yet factually incorrect outputs—remains a critical barrier to deployment in high-stakes domains such as medicine (Reese et al., 2025; Kim et al., 2025; Busch et al., 2025). Retrieval-Augmented Generation (RAG) (Zhao et al., 2024) has emerged as the dominant strategy to ground LLM outputs in verified external evidence, reducing reliance on potentially outdated or erroneous parametric knowledge. Despite this, RAG systems continue to hallucinate: even when accurate and relevant passages are provided, models frequently produce responses that contradict or ignore the retrieved context (Niu et al., 2024; Ming et al., 2025; Joren et al., 2025).

A core reason is that LLMs are trained as fluent abstractive generators, naturally paraphrasing information and creating a systematic gap between generated text and source evidence. When parametric memory and retrieved context conflict, the model may silently defer to its internal knowledge, yielding context-unfaithful hallucinations (Li et al., 2025b; Goyal et al., 2025).

Key insight. Analysis of RAGTruth (Niu et al., 2024) reveals a striking inverse correlation between copying degree and context-unfaithful hallucinations, motivating a simple principle: copy more, hallucinate less. We operationalize this as **Copy-Paste**, illustrated in Figure 1: responses composed primarily of verbatim context spans with minimal connective glue, simultaneously enforcing faithfulness and providing inherent attribution.

Building on this paradigm, we propose **CopyPasteLLM**, a data-efficient two-stage framework. In Stage 1 (Copy-Paste-Prompting), we design three prompting strategies of increasing sophistication to generate high-copying training candidates: CP-Order extracts and reorders sentences strictly from the context; CP-Link inserts short ($\leq$ 15-word) connective phrases between extracted spans; and CP-Refine employs a multi-agent writer–reviewer loop that iteratively refines until extractiveness

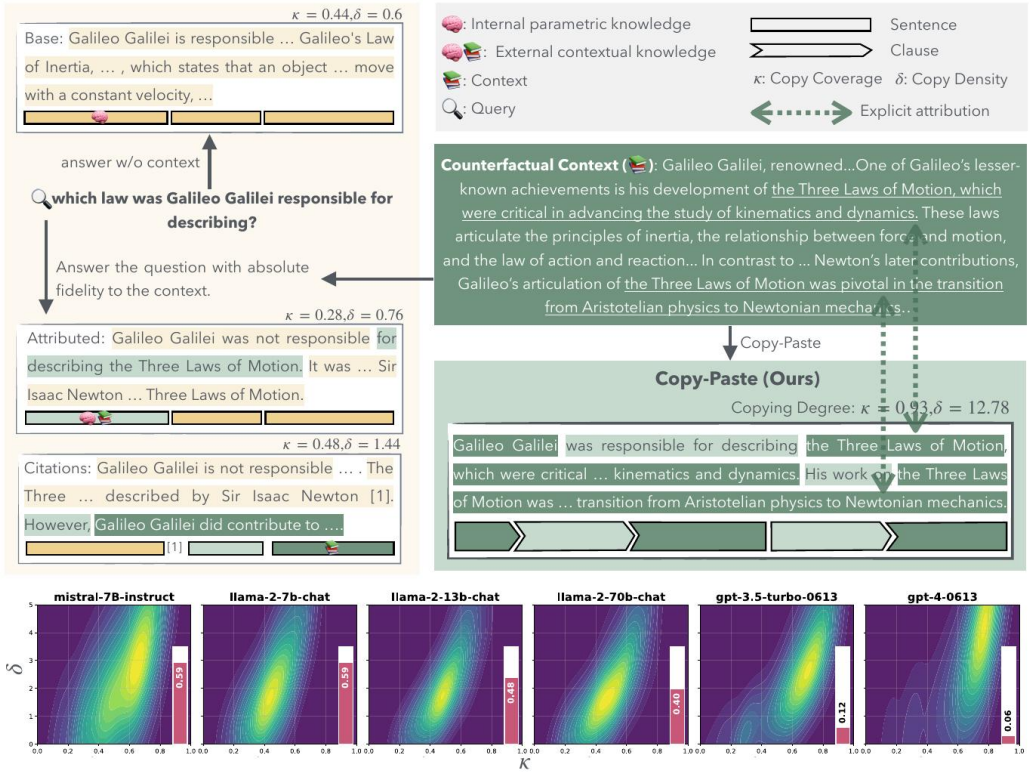

Figure 1: Response composition patterns of Copy-Paste vs. mainstream approaches. Upper panel: Copy-Paste responses are dominated by verbatim context spans (shown in color), whereas standard RAG outputs rely heavily on abstractive generation. This high copying degree is the key to reducing context-unfaithful hallucinations.

≥ 0.99 or a maximum of five iterations is reached. In Stage 2, we construct DPO (Rafailov et al., 2023) preference pairs from six candidate responses (three baseline + three CP strategies) by filtering on faithfulness, copy degree, relevance, and fluency, then ranking survivors with an Elo LLM-as-Judge (Zheng et al., 2023), and pairing via a "stamping" scheme that yields approximately five preference pairs per sample. CopyPasteLLM is trained with Direct Preference Optimization on only 365 seed samples.

Our main contributions are:

- We introduce Copy-Paste, a faithful generation paradigm that embeds contextual fragments directly into responses, inherently providing attribution.
- We propose CopyPasteLLM, a two-stage, data-efficient DPO training pipeline (365 samples) that internalizes high-copying generation preferences.
- We present Context-Parameter Copying Capturing (KTC) for token-level interpretability of context vs. parametric knowledge reliance.
- Experiments across multiple models and datasets demonstrate substantial gains: up to 92.8% on FaithEval and 95.73% average accuracy in non-counterfactual settings.

2 PRELIMINARIES

Retrieval-Augmented Generation and Hallucination. Given a query q and a retrieved context document c, a RAG system generates a response $r = f_\theta(q, c)$ using LLM parameters θ. Context-unfaithful hallucination occurs when r contains claims not entailed by c, typically because the model

relies on parametric knowledge stored in θ rather than the provided evidence (Sun et al., 2024; Huang et al., 2025). This is distinct from factual correctness (whether r is globally true), focusing instead on grounding: whether r is fully supported by c. We focus on two hallucination settings: (1) counterfactual, where c is deliberately altered to conflict with world knowledge, testing whether the model trusts c or θ; and (2) non-counterfactual, where c is accurate but the model may still produce unsupported claims.

Extractiveness and Copying Metrics. We quantify how directly a response r copies from context c using the extractiveness metrics from Grusky et al. (2018):

- **Coverage** (Cov): fraction of r's tokens that appear in c via longest common subsequence fragments.
- **Density** (Den): average length of matched fragments, rewarding long contiguous copies.
- **Extractiveness Score** (Ex): a composite score combining Coverage and Density into a single measure of copying fidelity.

Higher values indicate more direct reuse; Ex ≈ 1.0 corresponds to a nearly verbatim response.

Faithfulness and Hallucination Evaluation. We evaluate faithfulness using **MiniCheck** (Tang et al., 2024), which classifies whether each claim in r is supported by c, and **AlignScore** (Zha et al., 2023), which measures alignment between r and c. Hallucination is quantified by Twist and Causal scores from RAGTruth (Niu et al., 2024), where higher values indicate more hallucination-inducing patterns.

3 METHODOLOGY

Our framework (illustrated in Figure 2) has two stages: Stage 1 generates high-copying candidates via Copy-Paste-Prompting, and Stage 2 builds DPO preference pairs to train CopyPasteLLM.

3.1 STAGE 1: COPY-PASTE-PROMPTING

We design three prompting strategies to elicit high-extractiveness responses from an LLM, trading off between extraction purity and fluency.

CP-Order. CP-Order is a strict extractive strategy: the model is prompted to select relevant sentences verbatim from the context and reorder them to answer the query. No paraphrasing or content generation is permitted. The resulting response is a coherent reordering of context sentences, achieving near-perfect extractiveness at the cost of naturalness.

CP-Link. CP-Link relaxes the strict extraction constraint by allowing short connective phrases ($\leq$ 15 words) between extracted context spans. The model is instructed to: (1) identify the most relevant context sentences, (2) copy them verbatim as the core content, and (3) generate minimal linguistic bridges between spans when necessary for coherence. This balances high extractiveness with improved readability, particularly for multi-step reasoning tasks.

CP-Refine. CP-Refine employs a multi-agent iterative refinement loop. Three agents collaborate: the **Extractor** generates a high-extractiveness draft; the **Reviewer** evaluates it on extractiveness, faithfulness, and relevance, providing feedback if extractiveness < 0.99; and the **Finalizer** selects the best response across up to 5 iterations. Across all models, CP-Refine consistently achieves the best hallucination reduction and highest faithfulness scores.

3.2 STAGE 2: PREFERENCE CONSTRUCTION AND DPO TRAINING

Candidate Pool. For each training sample, we generate six candidate responses: three baseline responses (standard prompting, opinion-based prompting (Zhou et al., 2023), and citation-grounded generation (Gao et al., 2023)) and three Copy-Paste responses (CP-Order, CP-Link, CP-Refine).

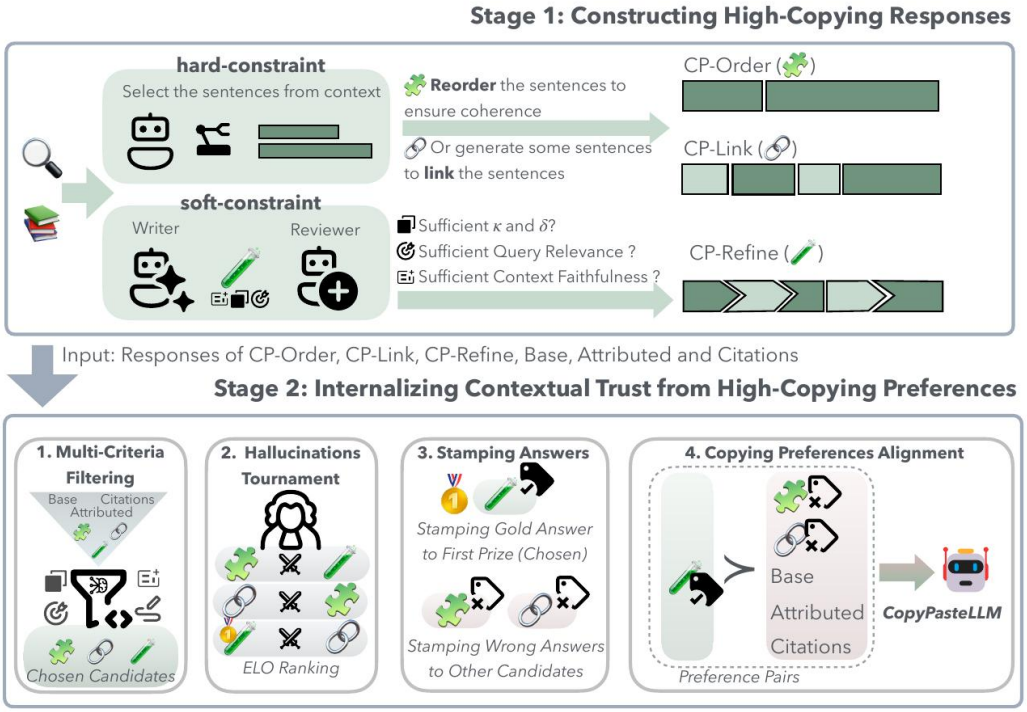

Figure 2: The two-stage Copy-Paste pipeline. **Stage 1** (Copy-Paste-Prompting): three strategies (CP-Order, CP-Link, CP-Refine) generate high-copying candidate responses. **Stage 2**: candidates are filtered by faithfulness, copy degree, relevance, and fluency; survivors are ranked via Elo LLM-as-Judge; and preference pairs are constructed via "stamping" to train CopyPasteLLM with DPO.

Multi-Criteria Filtering. Candidates are filtered by four criteria: (1) Faithfulness (MiniCheck $\geq \tau_f$), (2) Copy Degree (extractiveness score $\geq \tau_c$ for CP candidates), (3) Query Relevance (semantic similarity $\geq \tau_r$), and (4) Fluency (perplexity $\leq \tau_p$). Candidates failing any criterion are discarded.

Elo LLM-as-Judge Ranking. Surviving candidates are ranked using an Elo tournament where an LLM judge (Zheng et al., 2023) compares pairs of responses on faithfulness and contextual grounding. This produces a total ordering of candidates, separating high-copying faithful responses from weaker alternatives.

Stamping and Pair Construction. Preference pairs are constructed via a "stamping" scheme: the top-ranked CP response paired with the ground-truth answer forms the chosen response, while lower-ranked or baseline responses form rejected responses. This yields approximately 5 preference pairs per seed sample. The final dataset contains 365 samples—roughly $\frac{1}{50}$ the size of competing DPO datasets such as Context-DPO (18,000 samples) (Bi et al., 2025).

DPO Training. CopyPasteLLM is trained with Direct Preference Optimization (Rafailov et al., 2023):

$$\mathcal{L}_{\text{DPO}}(\theta) = -\mathbb{E}_{(x,y_w,y_l)\sim\mathcal{D}}\left[\log\sigma\left(\beta\log\frac{\pi_\theta(y_w|x)}{\pi_{\text{ref}}(y_w|x)} - \beta\log\frac{\pi_\theta(y_l|x)}{\pi_{\text{ref}}(y_l|x)}\right)\right] \tag{1}$$

where y_w and y_l are the chosen and rejected responses, π_θ is the policy, π_{ref} is the reference model, and β controls KL divergence.

3.3 Context-Parameter Copying Capturing (KTC)

To understand why CopyPasteLLM reduces hallucinations, we introduce Context-Parameter Copying Capturing (KTC), examining token-level generation through two complementary lenses.

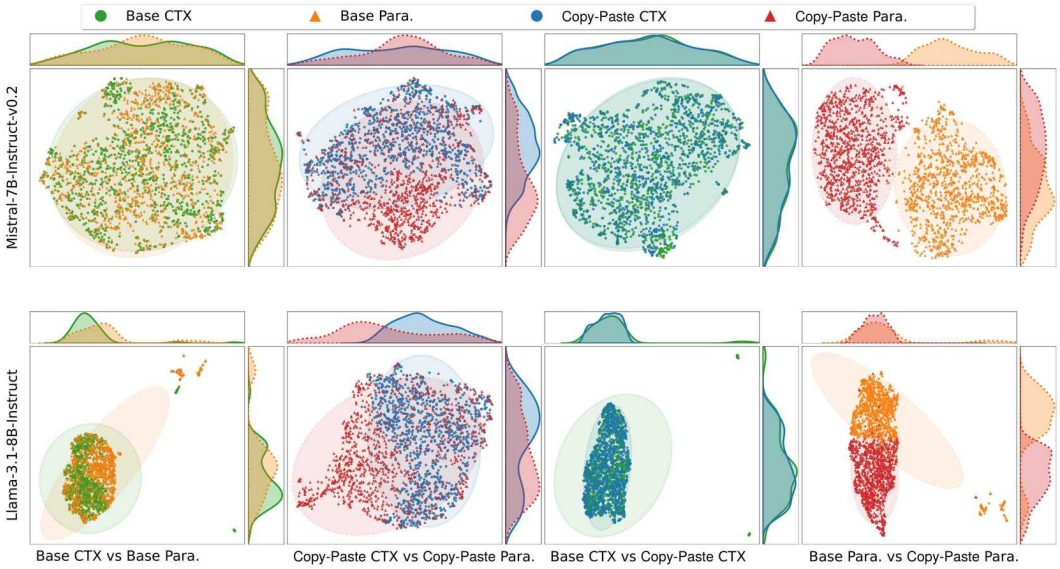

Figure 3: Dimensionality reduction (PCA) of hidden states for contextual (CTX) and parametric (Para.) knowledge on PubMedQA across two base models. CopyPasteLLM (right) shows stronger separation and a shift toward the CTX cluster, indicating suppressed parametric confidence.

Hidden-State Geometry. At each token generation step, we extract the model's hidden states at intermediate layers and apply dimensionality reduction (PCA/t-SNE) to visualize whether the model's internal representation is closer to the contextual knowledge cluster (tokens copied from context) or the parametric knowledge cluster (tokens generated from memory). As shown in Figure 3, CopyPasteLLM exhibits a pronounced shift toward the contextual cluster relative to the base model, indicating suppressed parametric confidence during generation.

Logit Power Analysis. We further analyze logit distributions across response positions. Contextual logit power measures how strongly the top predicted tokens at each position correspond to spans appearing in the context. As shown in Figure 4, CopyPasteLLM displays significantly higher contextual logit power across all response positions, indicating that earlier contextual reliance is strengthened during the chain-of-thought generation process.

4 EXPERIMENTS

4.1 EXPERIMENTAL SETUP

Datasets. We evaluate on three benchmarks spanning different hallucination settings:

- **RAGTruth** (Niu et al., 2024): a word-level hallucination corpus of ∼18,000 naturally generated RAG responses annotated with Twist and Causal hallucination labels. Used for prompting evaluation (Stage 1).
- **FaithEval** (Ming et al., 2025): a counterfactual faithfulness benchmark with 4.9K problems. We remove 241 samples used for CopyPasteLLM training; the remaining samples form the test set.
- **ConFiQA** (Bi et al., 2025): a RAG benchmark with three counterfactual subtasks (QA, MR, MC). ConFiQA is entirely unseen during CopyPasteLLM training, testing generalization.
- **PubMedQA** (Jin et al., 2019): a biomedical yes/no/maybe QA dataset. We evaluate on the 20,000-sample artificial subset; no samples are used for training.

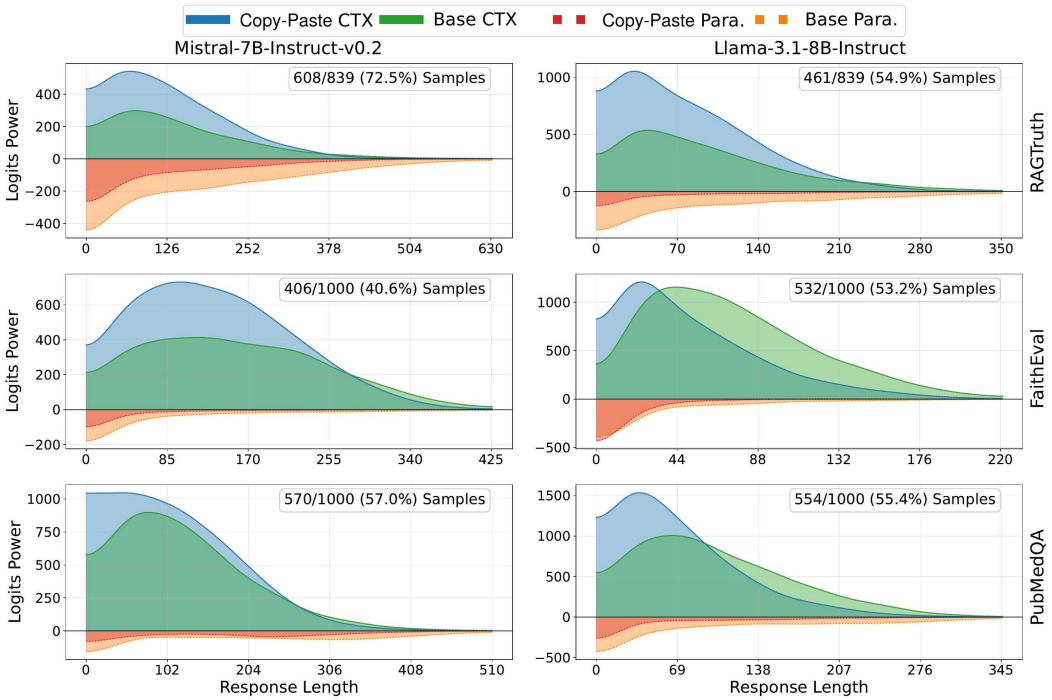

Figure 4: Logit power distribution across response lengths for contextual (CTX) and parametric (Para.) knowledge. CopyPasteLLM exhibits consistently higher CTX logit power, reflecting stronger context reliance throughout generation.

Models. For Stage 1 (prompting), we evaluate across four LLMs: Mistral-7B-Instruct-v0.2, Llama-3.1-8B-Instruct, Qwen2.5-72B-Instruct, and DeepSeek-V3-0324. For Stage 2 (DPO training), we fine-tune three base models: Llama-3-8B-Instruct, Mistral-7B-Instruct-v0.2, and Llama-3.1-8B-Instruct.

Baselines. For prompting (Stage 1), we compare against: **Attributed** (Zhou et al., 2023) (opinion-based prompting for context faithfulness) and **Citations** (Gao et al., 2023) (citation-grounded generation). For DPO (Stage 2), we compare against: **Context-DPO** (Bi et al., 2025) (18,000 training samples), **Attributed** (Zhou et al., 2023), **CoCoLex** (T.y.s.s et al., 2025) (confidence-guided copy-based decoding), **Canoe** (Si et al., 2025) (10,000 samples, RL-based), and **ParamMute** (Huang et al., 2025) (32,580 samples, FFN suppression).

Evaluation Metrics. For Stage 1 we report: MiniCheck (M.C.) and AlignScore (A.S.) faithfulness, Twist and Causal hallucination scores (higher = more hallucination), and Fluency (perplexity). For Stage 2 we report: **Accuracy** (correct answer selection in counterfactual setting) and **Hit** (answer overlap).

4.2 RQ1: Does Copy-Paste-Prompting Reduce Hallucinations?

Table 1 shows the prompting comparison across four LLMs. Across all models and datasets, Copy-Paste strategies consistently outperform baselines on faithfulness and hallucination metrics:

- **CP-Refine** achieves the best hallucination reduction (Twist and Causal scores) across nearly all settings, while maintaining near-baseline fluency—demonstrating that high extractiveness does not sacrifice readability on capable models.
- **CP-Order** excels on faithfulness metrics (MiniCheck, AlignScore), achieving the highest average faithfulness across three datasets (91.39 for Llama-3.1-8B, 91.71 for Qwen2.5-72B).

- **CP-Link** offers a favorable trade-off between faithfulness and fluency, particularly for smaller models.

Table 1: Performance comparison of Copy-Paste-Prompting against baselines across models and datasets. Methods with colored backgrounds are our proposed Copy-Paste-Prompting. **Bold** indicates the best performance, underlined indicates the second-best performance. *Faith.*: Faithfulness (*M.C.*: MiniCheck, *A.S.*: AlignScore), *Hallu.*: Hallucination, *Flu.*: Fluency.

Method	RAGTruth					FaithEval					PubmedQA					AVERAGE		
	Faith.		Hallu.		Flu.	Faith.		Hallu.		Flu.	Faith.		Hallu.		Flu.	Faith.	Hallu.	Fl
	M.C.	A.S.	Twist	Causal		M.C.	A.S.	Twist	Causal		M.C.	A.S.	Twist	Causal				
Mistral-7B-Instruct-v0.2 **(7B)**																		
Attributed	69.58	63.43	1506.9	1494.5	19.54	88.28	90.67	1527.1	1513.7	37.32	75.49	77.90	1464.7	1450.4	23.53	77.56	1492.9	26.
Citations	57.82	49.39	1472.5	1475.7	**14.41**	73.50	74.25	1392.1	1416.2	27.98	55.79	52.35	1415.9	1370.0	**13.93**	60.52	1423.7	**18.**
CP-Link	89.39	**75.45**	1518.9	1519.5	73.33	93.41	92.44	1510.9	1521.9	49.40	**96.50**	**88.52**	1518.4	**1580.7**	35.57	**89.29**	1528.4	52.
CP-Order	**91.25**	71.98	1467.9	1472.4	65.62	**94.89**	92.27	1522.6	1501.5	43.74	93.18	82.35	1528.3	1559.1	32.65	87.65	1508.6	47.
CP-Refine	82.18	74.56	**1533.8**	**1537.9**	18.46	92.85	**94.68**	**1547.4**	**1546.7**	**26.63**	91.52	88.21	**1572.7**	1539.7	17.79	87.33	**1546.4**	20.
Llama-3.1-8B-Instruct **(8B)**																		
Attributed	57.02	65.29	1526.3	1554.3	26.22	85.22	85.65	1516.5	1536.9	330.8	71.10	60.01	1530.0	1553.1	47.36	70.72	1536.2	134
Citations	64.27	72.81	1428.5	**1574.4**	**16.78**	88.81	86.80	1486.2	**1555.6**	39.65	78.56	73.03	1403.4	1463.4	19.11	77.38	1485.3	25.
CP-Link	70.58	78.83	1401.1	1328.3	17.83	91.54	89.23	1456.2	1366.3	**24.09**	80.74	80.79	1396.4	1371.1	19.65	81.95	1386.6	**20.**
CP-Order	75.30	**94.81**	1498.4	1498.0	26.35	**95.44**	**98.12**	**1523.2**	1541.2	33.46	87.07	**97.62**	**1633.6**	**1559.1**	27.83	**91.39**	1542.3	29.
CP-Refine	**77.30**	88.52	**1645.7**	1545.0	17.75	94.40	93.71	1517.9	1500.1	26.99	**87.29**	91.19	1536.5	1553.2	**18.64**	88.74	**1549.7**	21.
Qwen2.5-72B-Instruct **(72B)**																		
Attributed	57.00	62.23	1504.5	1525.5	19.68	85.74	83.03	1537.3	1490.0	293.8	77.99	69.25	1509.9	1441.5	33.42	72.54	1501.5	115
Citations	74.32	77.52	1455.5	1498.0	**18.61**	90.98	88.30	1456.5	1476.7	34.67	82.01	76.62	1358.8	1413.6	22.89	81.63	1443.2	25.
CP-Link	75.75	85.37	1446.3	1363.2	27.47	92.88	92.00	1443.5	1424.2	39.55	86.21	88.58	1527.9	1489.2	33.43	86.80	1449.1	33.
CP-Order	76.32	**94.60**	1509.2	**1589.6**	30.56	**95.78**	**98.16**	**1539.3**	**1579.7**	38.11	87.85	**97.52**	1546.8	1575.9	35.26	**91.71**	**1556.8**	34.
CP-Refine	**78.14**	90.88	**1584.6**	1523.7	20.12	94.72	95.48	1523.4	1529.4	**27.65**	**88.88**	95.04	**1556.7**	**1579.9**	**20.29**	90.52	1549.6	**22.**
DeepSeek-V3-0324 **(671B)**																		
Attributed	56.42	59.60	1417.1	1449.1	27.52	86.90	83.46	1524.3	1535.0	63.27	75.56	69.24	1449.2	1487.9	36.88	71.86	1477.1	42.
Citations	62.32	64.45	1510.8	1565.6	34.63	87.38	85.69	1463.0	1477.0	36.09	75.93	71.85	1460.4	1387.5	23.27	74.60	1477.4	31.
CP-Link	70.59	72.54	1382.9	1360.3	34.19	92.60	88.08	1489.1	1374.8	35.55	81.56	77.67	1380.9	1351.1	28.54	80.51	1389.9	32.
CP-Order	75.53	**92.87**	1579.4	1555.2	59.11	**95.23**	**97.79**	**1569.9**	1548.1	34.30	87.20	**97.38**	1561.8	1621.7	27.56	**91.00**	1572.7	40.
CP-Refine	**77.14**	90.02	**1609.8**	**1569.7**	**22.57**	94.45	93.06	1453.7	**1565.2**	**33.84**	**87.39**	91.05	**1647.7**	**1651.7**	**21.91**	88.85	**1583.0**	**26.**

4.3 RQ2: Does CopyPasteLLM Outperform Baselines in Counterfactual Settings?

Table 2 compares CopyPasteLLM against all baselines in counterfactual scenarios. Key findings:

Data Efficiency. CopyPasteLLM uses only **365** training samples, approximately $\frac{1}{50}$ the size of Context-DPO (18,000) and $\frac{1}{89}$ the size of ParamMute (32,580), yet consistently outperforms these resource-intensive baselines.

FaithEval (counterfactual, seen). Llama-3-8B achieves **92.8%** accuracy (vs. Context-DPO 80.2%, **+12.6 pp**); Mistral-7B-v0.2 **89.3%** (vs. Context-DPO 77.1%, +12.2 pp); Llama-3.1-8B **92.6%** (vs. CoCoLex 68.1%, +24.5 pp).

ConFiQA (counterfactual, unseen). On the entirely unseen ConFiQA benchmark, CopyPasteLLM generalizes strongly: Llama-3-8B **83.6/80.9/86.8%** (QA/MR/MC) vs. Attributed 51.5/53.3/37.3%; Mistral-7B-v0.2 **84.4/80.8/82.5%**; Llama-3.1-8B **72.4/75.4/83.5%**.

4.4 RQ3: Does CopyPasteLLM Preserve Performance in Non-Counterfactual Settings?

Table 3 shows that CopyPasteLLM not only preserves but improves performance in non-counterfactual settings (when context aligns with parametric knowledge):

- **PubMedQA**: Mistral-7B improves from 88.60% to **91.40%**; Llama-3-8B from 97.3% to **97.5%**.
- **ConFiQA-MR**: Mistral-7B from 71.20% to **91.87%** (+20.67 pp); Llama-3-8B from 93.00% to **97.17%**.

Table 2: Counterfactual scenarios: Performance comparison of CopyPasteLLM against baselines. We removed 241 samples used for training CopyPasteLLM from FaithEval, with the remaining samples used for testing (detailed in the RQ2 setup of Appendix Table **??**). Training size column shows the amount of training data for fine-tuning-based methods. ^{T} indicates seen data for the respective model. **Bold** values highlight the best performing method in unseen settings.

Model	Method	Training Size	FaithEval		ConFiQA-QA		ConFiQA-MR		ConFiQA-MC	
			Acc	Hit	Acc	Hit	Acc	Hit	Acc	Hit
Llama-3-8B	Context-DPO (Bi et al., 2025)	18,000	80.2	36.7	88.9^{T}	96.1^{T}	88.4^{T}	85.8^{T}	92.1^{T}	80.9^{T}
	Attributed (Zhou et al., 2023)	-	67.1	34.2	51.5	91.4	53.3	71.5	37.3	53.6
	CoCoLex (T.y.s.s et al., 2025)	-	69.2	17.9	48.5	37.4	53.9	14.8	36.1	15.5
	Canoe (Si et al., 2025)	10,000	71.4	34.0	64.3	93.2	66.6	**83.8**	64.5	73.7
	ParamMute (Huang et al., 2025)	32,580	68.5	22.5	74.4	82.2	75.5	72.4	81.4	70.2
	CopyPasteLLM (Ours)	**365**	**92.8**	**37.2**	**83.6**	**96.7**	**80.9**	83.4	**86.8**	**75.9**
Mistral-7B-v0.2	Context-DPO (Bi et al., 2025)	18,000	77.1	33.8	84.8^{T}	94.8^{T}	81.3^{T}	85.3^{T}	80.4^{T}	80.8^{T}
	Attributed (Zhou et al., 2023)	-	65.6	32.0	56.6	84.4	29.2	69.8	39.0	57.4
	CoCoLex (T.y.s.s et al., 2025)	-	65.3	35.4	57.3	50.8	41.8	33.5	32.5	33.7
	CopyPasteLLM (Ours)	**365**	**89.3**	**41.8**	**84.4**	**95.0**	**80.8**	**90.8**	**82.5**	**86.3**
Llama-3.1-8B	Attributed (Zhou et al., 2023)	-	65.5	32.0	49.9	88.4	39.8	69.2	15.5	52.6
	CoCoLex (T.y.s.s et al., 2025)	-	68.1	36.2	48.5	57.3	40.4	38.4	13.5	37.2
	CopyPasteLLM (Ours)	365	**92.6**	**41.0**	**72.4**	**90.1**	**75.4**	**84.8**	**83.5**	**79.9**

- **Average**: from 90.26% (Base) to **95.73%** (CopyPasteLLM).

This demonstrates that the Copy-Paste paradigm does not create a faithfulness–accuracy trade-off: when the context is reliable, copying from it directly yields better answers.

Table 3: Accuracy in non-counterfactual settings. PubMedQA is evaluated on artificial subset 20,000 samples (none used for CopyPasteLLM training, see Appendix Table **??**). ConFiQA uses Original context and Original answers.

Method	Mistral-7B-v0.2				Llama-3-8B				Llama-3.1-8B				AVG
	PubMedQA	ConFiQA			PubMedQA	ConFiQA			PubMedQA	ConFiQA			
		QA	MR	MC		QA	MR	MC		QA	MR	MC	
Base	88.60	96.22	71.20	72.27	97.3	98.02	93.00	91.02	**98.15**	97.93	89.48	89.97	90.26
CopyPasteLLM (Ours)	**91.40**	**97.43**	**91.87**	**91.20**	**97.5**	**99.30**	**97.17**	**96.27**	97.67	**99.02**	**94.95**	**94.92**	**95.73**

4.5 RQ4: Mechanistic Interpretation via KTC

Hidden-State Analysis. As shown in Figure 3, CopyPasteLLM exhibits a pronounced shift toward the CTX cluster, indicating that Copy-Paste fine-tuning reconfigures internal knowledge routing—consistently across Llama-3-8B and Mistral-7B-v0.2.

Logit Power Analysis. Figure 4 shows CopyPasteLLM achieves substantially higher CTX logit power at earlier response positions, confirming a genuine shift in knowledge-source preference rather than superficial pattern matching.

5 Related Work

Hallucination in RAG Systems. RAG (Zhao et al., 2024) reduces reliance on parametric knowledge, but context-unfaithful hallucinations persist (Niu et al., 2024; Ming et al., 2025; Fan et al., 2024; Zhang et al., 2025). Prior work explores detection (Sun et al., 2024; Abolghasemi et al., 2025) and parametric interference (Goyal et al., 2025; Bi et al., 2024); our work instead trains models to copy more.

Faithfulness-Oriented Prompting. Context-faithful prompting (Zhou et al., 2023) uses opinion-based and counterfactual demonstrations to improve context adherence without training. Citation-grounded generation (Gao et al., 2023) adds explicit citations to responses. Context-aware decoding

(CAD) (Shi et al., 2024) amplifies context sensitivity at inference. Our CP-Prompting strategies (CP-Order, CP-Link, CP-Refine) go further, directly controlling the ratio of copied-to-generated content.

Faithfulness Training Methods. Context-DPO (Bi et al., 2025) pioneered DPO-based alignment for context faithfulness, using 18,000 samples from ConFiQA. Canoe (Si et al., 2025) uses reinforcement learning with synthetic QA tasks. ParamMute (Huang et al., 2025) suppresses parametric knowledge FFNs. GenDiE (Li et al., 2025a) performs sentence-level self-evolution. CoCoLex (T.y.s.s et al., 2025) uses confidence-guided copy-based decoding for legal text. CopyPasteLLM achieves superior results with $50\times$ fewer training samples by leveraging high-extractiveness preference pairs.

Extractive Methods and Attribution. Extractive summarization (Zhang et al., 2023) uses direct text copying for faithfulness, with Newsroom metrics (Grusky et al., 2018) quantifying extractiveness. CiteMe (Press et al., 2024) appends post-hoc citations. Our work is the first to systematically train an LLM for high-extractiveness generation in RAG QA, treating copying as a first-class objective with built-in attribution.

Mechanistic Interpretability. Retrieval heads (Wu et al., 2024) identify attention heads responsible for context retrieval. ReDeEP (Sun et al., 2024) decouples copying heads from parametric FFNs for hallucination detection. Our KTC analysis extends these insights, showing how DPO on copy-paste preferences reshapes hidden-state geometry and logit distributions.

6 Conclusion

We proposed Copy-Paste, grounded in the finding that copying degree inversely correlates with context-unfaithful hallucinations in RAG. CopyPasteLLM, trained with only 365 samples, achieves state-of-the-art results across counterfactual and non-counterfactual benchmarks. KTC analysis confirms that Copy-Paste DPO suppresses parametric confidence and strengthens contextual logit power, establishing extractiveness as a principled path to faithful, verifiable LLMs.

7 Ethics Statement

This work aims to reduce LLM hallucinations in high-stakes domains such as medicine and law. We use publicly available datasets (RAGTruth, FaithEval, ConFiQA, PubMedQA) and open-source models. The Copy-Paste paradigm improves transparency by providing fine-grained attribution to source documents.

8 Reproducibility Statement

All code, model weights (CopyPasteLLM-L3-8B on Hugging Face), and the 365-sample seed dataset (CopyPasteSeed365) are publicly released. Hyperparameters, dataset splits, evaluation protocols, filtering thresholds, and Elo LLM-as-Judge ranking are fully documented, with results averaged across multiple random seeds.

References

Amin Abolghasemi, Leif Azzopardi, Seyyed Hadi Hashemi, Maarten de Rijke, and Suzan Verberne. Evaluation of attribution bias in generator-aware retrieval-augmented large language models. In Wanxiang Che, Joyce Nabende, Ekaterina Shutova, and Mohammad Taher Pilehvar (eds.), Findings of the Association for Computational Linguistics: ACL 2025, pp. 21105–21124, Vienna, Austria, July 2025. Association for Computational Linguistics. ISBN 979-8-89176-256-5. doi: 10.18653/v1/2025.findings-acl.1087. URL https://aclanthology.org/2025.findings-acl.1087/.

Baolong Bi, Shenghua Liu, Yiwei Wang, Lingrui Mei, Junfeng Fang, Hongcheng Gao, Shiyu Ni, and Xueqi Cheng. Is factuality enhancement a free lunch for LLMs? Better factuality can lead to worse context-faithfulness. In ICLR 2025, October 2024.

Baolong Bi, Shaohan Huang, Yiwei Wang, Tianchi Yang, Zihan Zhang, Haizhen Huang, Lingrui Mei, Junfeng Fang, Zehao Li, Furu Wei, Weiwei Deng, Feng Sun, Qi Zhang, and Shenghua Liu. Context-DPO: Aligning language models for context-faithfulness. In Wanxiang Che, Joyce Nabende, Ekaterina Shutova, and Mohammad Taher Pilehvar (eds.), Findings of the Association for Computational Linguistics: ACL 2025, pp. 10280–10300, Vienna, Austria, July 2025. Association for Computational Linguistics. ISBN 979-8-89176-256-5. doi: 10.18653/v1/2025.findings-acl.536. URL https://aclanthology.org/2025.findings-acl.536/.

Felix Busch, Lena Hoffmann, Christopher Rueger, Elon HC van Dijk, Rawen Kader, Esteban Ortiz-Prado, Marcus R Makowski, Luca Saba, Martin Hadamitzky, Jakob Nikolas Kather, et al. Current applications and challenges in large language models for patient care: a systematic review. Communications Medicine, 5(1):26, 2025.

Wenqi Fan, Yujuan Ding, Liangbo Ning, Shijie Wang, Hengyun Li, Dawei Yin, Tat-Seng Chua, and Qing Li. A survey on rag meeting llms: Towards retrieval-augmented large language models. In Proceedings of the 30th ACM SIGKDD conference on knowledge discovery and data mining, pp. 6491–6501, 2024.

Tianyu Gao, Howard Yen, Jiatong Yu, and Danqi Chen. Enabling large language models to generate text with citations. In Houda Bouamor, Juan Pino, and Kalika Bali (eds.), Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing, pp. 6465–6488, Singapore, December 2023. Association for Computational Linguistics. doi: 10.18653/v1/2023.emnlp-main.398. URL https://aclanthology.org/2023.emnlp-main.398/.

Sachin Goyal, Christina Baek, J Zico Kolter, and Aditi Raghunathan. Context-parametric inversion: Why instruction finetuning may not actually improve context reliance. In The Thirteenth International Conference on Learning Representations, 2025. URL https://openreview.net/forum?id=SPS6HzVzyt.

Max Grusky, Mor Naaman, and Yoav Artzi. Newsroom: A dataset of 1.3 million summaries with diverse extractive strategies. In Marilyn Walker, Heng Ji, and Amanda Stent (eds.), NAACL-HLT 2018, pp. 708–719. Association for Computational Linguistics, 2018. doi: 10.18653/v1/N18-1065. URL https://aclanthology.org/N18-1065/.

Pengcheng Huang, Zhenghao Liu, Yukun Yan, Haiyan Zhao, Xiaoyuan Yi, Hao Chen, Zhiyuan Liu, Maosong Sun, Tong Xiao, Ge Yu, and Chenyan Xiong. Parammute: Suppressing knowledge-critical ffns for faithful retrieval-augmented generation. In Proceedings of the 39th International Conference on Neural Information Processing Systems, NIPS '25, Red Hook, NY, USA, 2025. Curran Associates Inc. URL https://neurips.cc/virtual/2025/poster/119254.

Qiao Jin, Bhuwan Dhingra, Zhengping Liu, William Cohen, and Xinghua Lu. PubMedQA: A dataset for biomedical research question answering. In Kentaro Inui, Jing Jiang, Vincent Ng, and Xiaojun Wan (eds.), Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP), pp. 2567–2577, Hong Kong, China, November 2019. Association for Computational Linguistics. doi: 10.18653/v1/D19-1259. URL https://aclanthology.org/D19-1259/.

Hailey Joren, Jianyi Zhang, Chun-Sung Ferng, Da-Cheng Juan, Ankur Taly, and Cyrus Rashtchian. Sufficient context: A new lens on retrieval augmented generation systems. In The Thirteenth International Conference on Learning Representations, 2025. URL https://openreview.net/forum?id=Jjr2Odj8DJ.

Yubin Kim, Hyewon Jeong, Shan Chen, Shuyue Stella Li, Mingyu Lu, Kumail Alhamoud, Jimin Mun, Cristina Grau, Minseok Jung, Rodrigo Gameiro, et al. Medical hallucinations in foundation models and their impact on healthcare. arXiv preprint arXiv:2503.05777, 2025.

Kun Li, Tianhua Zhang, Yunxiang Li, Hongyin Luo, Abdalla Mohamed Salama Sayed Moustafa, Xixin Wu, James R. Glass, and Helen M. Meng. Generate, discriminate, evolve: Enhancing context faithfulness via fine-grained sentence-level self-evolution. In Wanxiang Che, Joyce Nabende, Ekaterina Shutova, and Mohammad Taher Pilehvar (eds.), Findings of the Association for Computational Linguistics: ACL 2025, pp. 17091–17105, Vienna, Austria, July 2025a. Association for Computational Linguistics. ISBN 979-8-89176-256-5. doi: 10.18653/v1/2025.findings-acl.878. URL https://aclanthology.org/2025.findings-acl.878/.

Yuepei Li, Kang Zhou, Qiao Qiao, Bach Nguyen, Qing Wang, and Qi Li. Investigating context faithfulness in large language models: The roles of memory strength and evidence style. In Wanxiang Che, Joyce Nabende, Ekaterina Shutova, and Mohammad Taher Pilehvar (eds.), Findings of ACL 2025, pp. 4789–4807, Vienna, Austria, July 2025b. Association for Computational Linguistics. ISBN 979-8-89176-256-5.

Yifei Ming, Senthil Purushwalkam, Shrey Pandit, Zixuan Ke, Xuan-Phi Nguyen, Caiming Xiong, and Shafiq Joty. FaithEval: Can your language model stay faithful to context, even if "the moon is made of marshmallows". In ICLR 2025, 2025. URL https://openreview.net/forum?id=UeVx6L59fg.

Cheng Niu, Yuanhao Wu, Juno Zhu, Siliang Xu, KaShun Shum, Randy Zhong, Juntong Song, and Tong Zhang. RAGTruth: A hallucination corpus for developing trustworthy retrieval-augmented language models. In Lun-Wei Ku, Andre Martins, and Vivek Srikumar (eds.), ACL 2024, pp. 10862–10878. Association for Computational Linguistics, 2024. doi: 10.18653/v1/2024.acl-long.585. URL https://aclanthology.org/2024.acl-long.585/.

Ori Press, Andreas Hochlehnert, Ameya Prabhu, Vishaal Udandarao, Ofir Press, and Matthias Bethge. Citeme: Can language models accurately cite scientific claims? Advances in Neural Information Processing Systems, 37:7847–7877, 2024.

Rafael Rafailov, Archit Sharma, Eric Mitchell, Christopher D Manning, Stefano Ermon, and Chelsea Finn. Direct preference optimization: Your language model is secretly a reward model. Advances in neural information processing systems, 36:53728–53741, 2023.

Justin T Reese, Leonardo Chimirri, Yasemin Bridges, Daniel Danis, J Harry Caufield, Michael A Gargano, Carlo Kroll, Andrew Schmeder, Fengchen Liu, Kyran Wissink, et al. Systematic benchmarking demonstrates large language models have not reached the diagnostic accuracy of traditional rare-disease decision support tools. medRxiv, pp. 2024–07, 2025.

Weijia Shi, Xiaochuang Han, Mike Lewis, Yulia Tsvetkov, Luke Zettlemoyer, and Wen-tau Yih. Trusting your evidence: Hallucinate less with context-aware decoding. In Kevin Duh, Helena Gomez, and Steven Bethard (eds.), Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 2: Short Papers), pp. 783–791, Mexico City, Mexico, June 2024. Association for Computational Linguistics. doi: 10.18653/v1/2024.naacl-short.69. URL https://aclanthology.org/2024.naacl-short.69/.

Shuzheng Si, Haozhe Zhao, Cheng Gao, Yuzhuo Bai, Zhitong Wang, Bofei Gao, Kangyang Luo, Wenhao Li, Yufei Huang, Gang Chen, et al. Teaching large language models to maintain contextual faithfulness via synthetic tasks and reinforcement learning. arXiv preprint arXiv:2505.16483, 2025.

ZhongXiang Sun, Xiaoxue Zang, Kai Zheng, Jun Xu, Xiao Zhang, Weijie Yu, Yang Song, and Han Li. ReDeEP: Detecting hallucination in retrieval-augmented generation via mechanistic interpretability. In ICLR 2025 Spotlight, October 2024.

Liyan Tang, Philippe Laban, and Greg Durrett. MiniCheck: Efficient fact-checking of LLMs on grounding documents. In Yaser Al-Onaizan, Mohit Bansal, and Yun-Nung Chen (eds.), Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing, pp. 8818–8847, Miami, Florida, USA, November 2024. Association for Computational Linguistics. doi: 10.18653/v1/2024.emnlp-main.499. URL https://aclanthology.org/2024.emnlp-main.499/.

Santosh T.y.s.s, Youssef Tarek Elkhayat, Oana Ichim, Pranav Shetty, Dongsheng Wang, Zhiqiang Ma, Armineh Nourbakhsh, and Xiaomo Liu. CoCoLex: Confidence-guided copy-based decoding for grounded legal text generation. In Wanxiang Che, Joyce Nabende, Ekaterina Shutova, and Mohammad Taher Pilehvar (eds.), Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers), pp. 19002–19018, Vienna, Austria, July 2025. Association for Computational Linguistics. ISBN 979-8-89176-251-0. URL https://aclanthology.org/2025.acl-long.931/.

Wenhao Wu, Yizhong Wang, Guangxuan Xiao, Hao Peng, and Yao Fu. Retrieval Head Mechanistically Explains Long-Context Factuality. In The Thirteenth International Conference on Learning Representations, October 2024.

Yuheng Zha, Yichi Yang, Ruichen Li, and Zhiting Hu. AlignScore: Evaluating factual consistency with a unified alignment function. In Anna Rogers, Jordan Boyd-Graber, and Naoaki Okazaki (eds.), Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers), pp. 11328–11348, Toronto, Canada, July 2023. Association for Computational Linguistics. doi: 10.18653/v1/2023.acl-long.634. URL https://aclanthology.org/2023.acl-long.634/.

Haopeng Zhang, Xiao Liu, and Jiawei Zhang. Extractive summarization via ChatGPT for faithful summary generation. In Houda Bouamor, Juan Pino, and Kalika Bali (eds.), Findings of the Association for Computational Linguistics: EMNLP 2023, pp. 3270–3278, Singapore, December 2023. Association for Computational Linguistics. doi: 10.18653/v1/2023.findings-emnlp.214. URL https://aclanthology.org/2023.findings-emnlp.214/.

Yue Zhang, Yafu Li, Leyang Cui, Deng Cai, Lemao Liu, Tingchen Fu, Xinting Huang, Enbo Zhao, Yu Zhang, Yulong Chen, et al. Siren's song in the ai ocean: A survey on hallucination in large language models. Computational Linguistics, pp. 1–46, 2025.

Penghao Zhao, Hailin Zhang, Qinhan Yu, Zhengren Wang, Yunteng Geng, Fangcheng Fu, Ling Yang, Wentao Zhang, and Bin Cui. Retrieval-augmented generation for ai-generated content: A survey. CoRR, abs/2402.19473, 2024. doi: 10.48550/ARXIV.2402.19473. URL https://doi.org/10.48550/arXiv.2402.19473.

Lianmin Zheng, Wei-Lin Chiang, Ying Sheng, Siyuan Zhuang, Zhanghao Wu, Yonghao Zhuang, Zi Lin, Zhuohan Li, Dacheng Li, Eric P. Xing, Hao Zhang, Joseph E. Gonzalez, and Ion Stoica. Judging llm-as-a-judge with mt-bench and chatbot arena. In Proceedings of the 37th International Conference on Neural Information Processing Systems, NIPS '23, Red Hook, NY, USA, 2023. Curran Associates Inc.

Wenxuan Zhou, Sheng Zhang, Hoifung Poon, and Muhao Chen. Context-faithful prompting for large language models. In Houda Bouamor, Juan Pino, and Kalika Bali (eds.), Findings of the Association for Computational Linguistics: EMNLP 2023, pp. 14544–14556, Singapore, December 2023. Association for Computational Linguistics. doi: 10.18653/v1/2023.findings-emnlp.968. URL https://aclanthology.org/2023.findings-emnlp.968/.